MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, FET, SWITCH, DIGITAL, KEY CODE GDP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic FET switch driver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washinyton, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.80 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GDP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/178-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

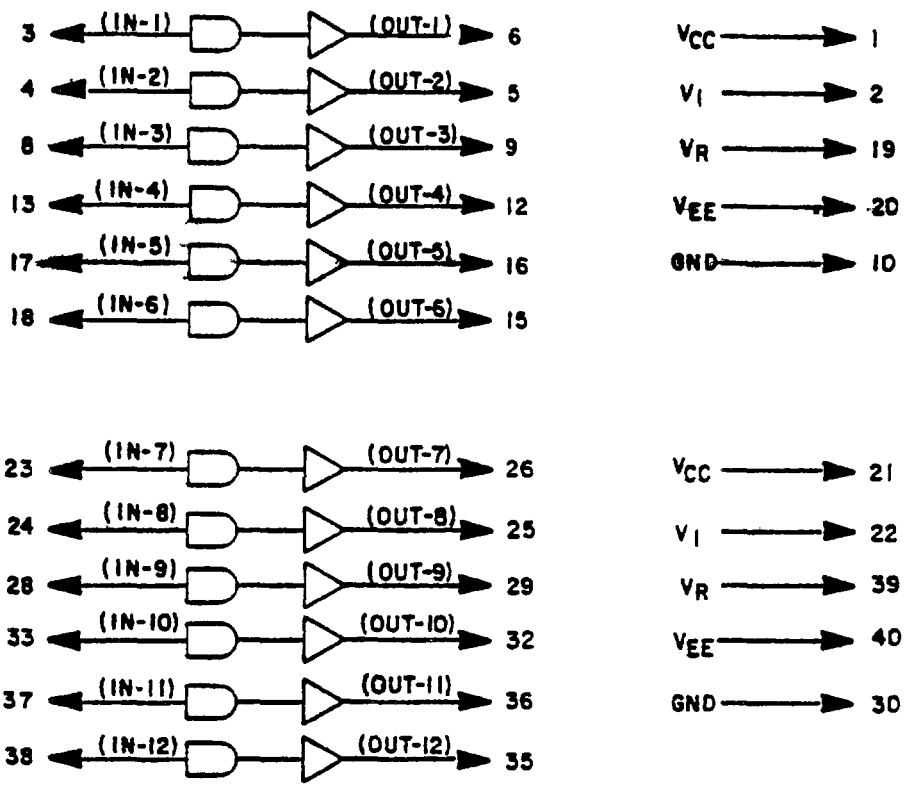
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. V_{CC} = Output positive reference supply voltage.
2. V_{EE} = Output negative reference supply voltage.
3. V_I = Input logic supply voltage.
4. V_R = Input logic reference voltage.
5. IN = Input.
6. OUT = Output.
7. Pins 7, 11, 14, 27, 31, and 34 are unused pins.

FIGURE 1. GDP module function diagram, FET switch driver.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Stress (maximum voltage ratings)	II	S	--	--	--	--	--	--	--	--	--
C1-C7	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D12	Input current, high level	IV	X	-1	10	-2	10	-4	20	-5	20	μA
E1-E12	Input current, low level	IV	X	0	1	0	1	0	1	0	1	mA
F1-F12	Output voltage, high level	V	X	7.5	10.0	7.5	10.0	7.5	10.0	7.5	10.0	V
G1-G12	Output voltage, low level	V	X	-17.5	-20.0	-17.5	-20.0	-17.5	-20.0	-17.5	-20.0	V
H1	Power supply current, high level I_{CC} (V_{CC} = 10 V)	VI	X	0	45	0	45	0	45	0	45	mA
H2	Power supply current, high level I_{EE} (V_{EE} = -20 V)	VI	X	0	-24	0	-24	0	-24	0	-24	mA
H3	Power supply current, high level I_I (V_I = 5 V)	VI	X	4	24	4	24	4	24	4	24	mA
H4	Power supply current, high level I_R (V_R = 0 V)	VI	X	-1	-54	-1	-54	-1	-54	-1	-54	mA
J1	Power supply current, low level I_{CC} (V_{CC} = 10 V)	VI	X	0	3	0	3	0	3	0	3	mA
J2	Power supply current, low level I_{EE} (V_{EE} = -20 V)	VI	X	-3	-30	-3	-30	-3	-30	-3	-30	mA
J3	Power supply current, low level I_I (V_I = 5 V)	VI	X	3	36	3	36	3	36	3	36	mA
J4	Power supply current, low level I_R (V_R = 0 V)	VI	X	0.0	-9.6	0.0	-9.6	0.0	-9.6	0.0	-9.6	mA
K1-K12	Propagation delay time (t_{PLH})	VII	S	50	450	50	500	50	500	50	500	ns
L1-L12	Propagation delay time (t_{PHL})	VII	S	50	450	50	500	50	500	50	500	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M12	Output crosstalk, low level	VIII	Q	0	1	0	1	0	1	0	1	V
N1-N12	Output crosstalk, high level	VIII	Q	0	800	0	800	0	800	0	800	mV
P1	Power supply transient amplitude (V_I)	IX	Q	0	400	0	400	0	400	0	400	mV
P2,P3	Power supply transient amplitude (V_{CC}, V_{EE})	IX	Q	0	2	0	2	0	2	0	2	V
Q1-Q12	Input capacitance	X	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through Q shall be performed in any order with the following exceptions. If test conditions A or C are part of a sequence of tests, the order of testing shall be as follows: A, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Q in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through L shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through L shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through Q shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

*6.3 Cross reference of classification. Military Specification MIL-M-28787/178 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658680. Part number M28787/178-1 supersedes part numbers 2658680 and M28787/178.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage (typical)	
V_{CC}	10 V dc
V_{EE}	-20 V dc
V_1	5 V dc
V_R	0 V dc

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Power supply maximum ratings		
$V_{CC}-V_{EE}$	36 V	
$V_{CC}-V_R$	15 V	
V_R-V_{EE}	25 V	
V_1-V_R	8 V	
V_1-V_{in}	8 V	
$V_{in}-V_R$	6 V	
Power supply current output at	**High level**	**Low level**
I_{CC} (V_{CC} = 10 V dc)	45 mA maximum	3 mA maximum
I_{EE} (V_{EE} = -20 V dc)	-24 mA maximum	-30 mA maximum
I_1 (V_1 = 5 V dc)	24 mA maximum	36 mA maximum
I_R (V_R = 0 V dc)	-54 mA maximum	-9.6 mA maximum
Input voltage		
Low level	0.8 V maximum	
High level	2.0 V maximum	
Output voltage		
Low level	V_{EE} +2.5 V maximum	
High level	V_{CC} -2.5 V maximum	
Propagation delay time		
t_{PLH}	500 ns maximum	
t_{PHL}	500 ns maximum	

6.5 Module function. This module contains 12 driver circuits designed to provide the level shifting and amplification required to interface low-level logic output (0.8 to 2.0 V) with high-level (to 30 V peak-to-peak) FET switches. Each driver circuit has an independent input (in) and output (out).

6.5.1 Typical driver circuit operation. With the input logic reference (V_R) at 0 V, the output negative reference supply (V_{EE}) may be set between -5 and -20 V. The output positive reference supply (V_{CC}) may be set up to 30 V above V_{EE}. A positive high level at the input provides a high level at the output and a low level at the input provides a low level at the output.

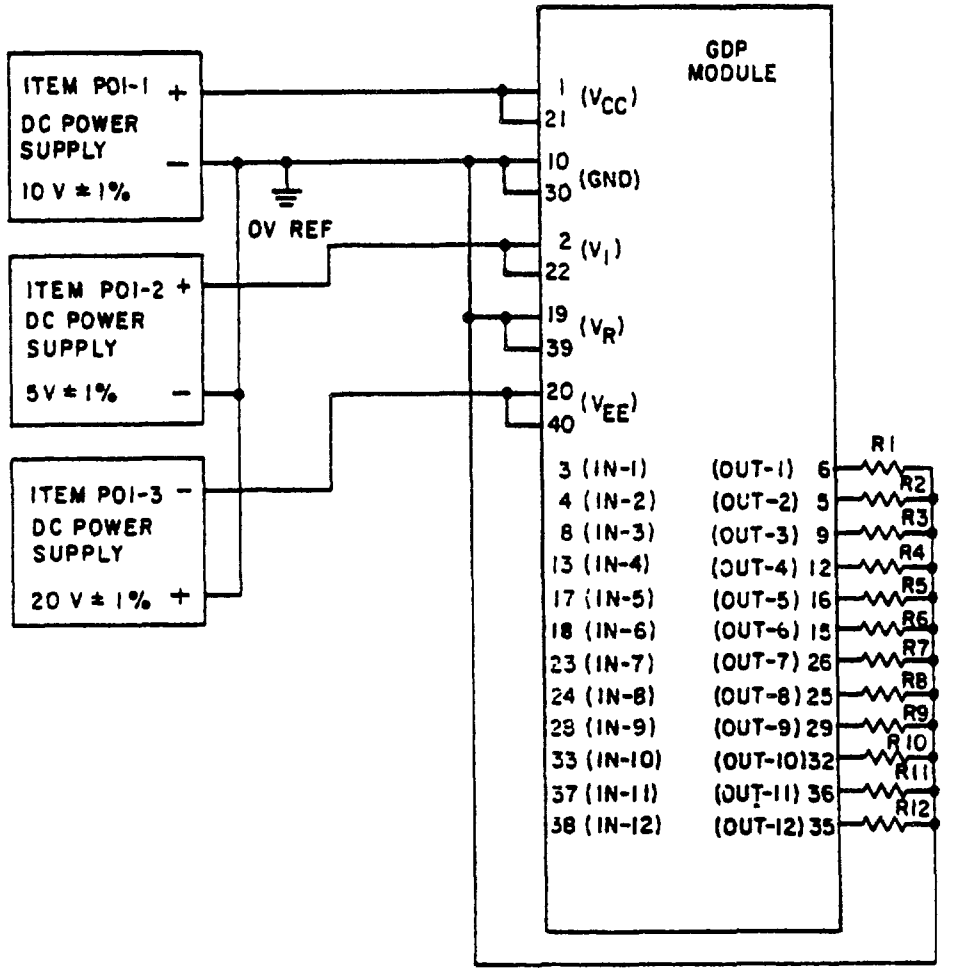
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE: R1-R12 = 3.57 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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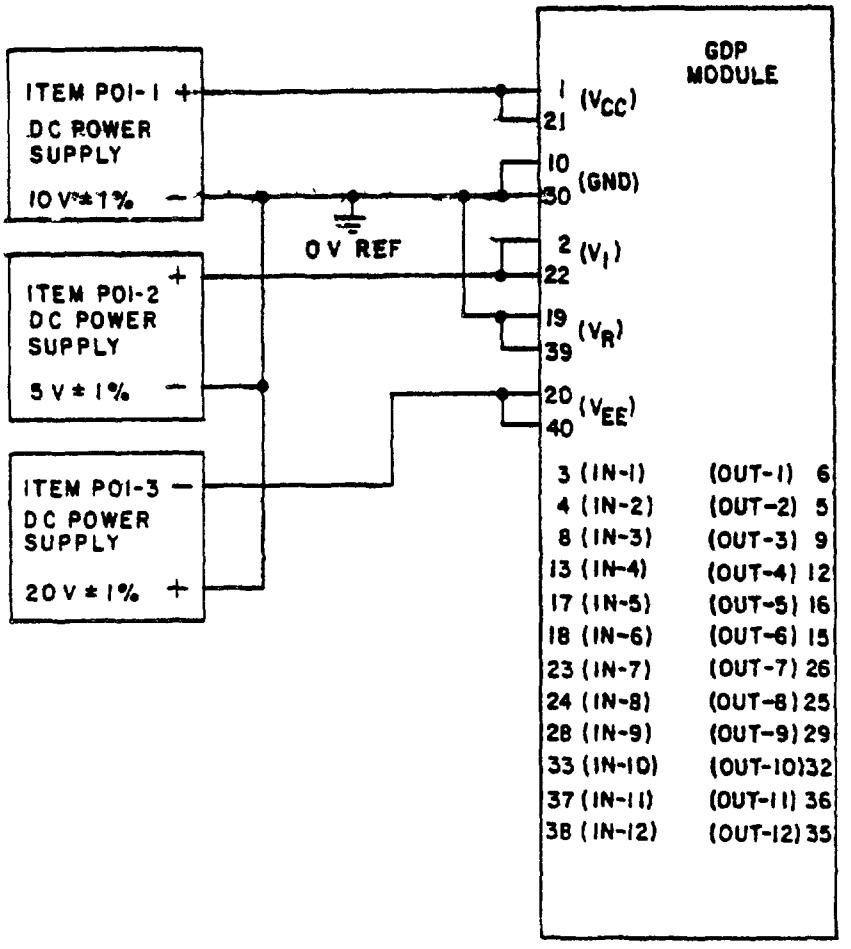


FIGURE 3. Vibration test circuit.

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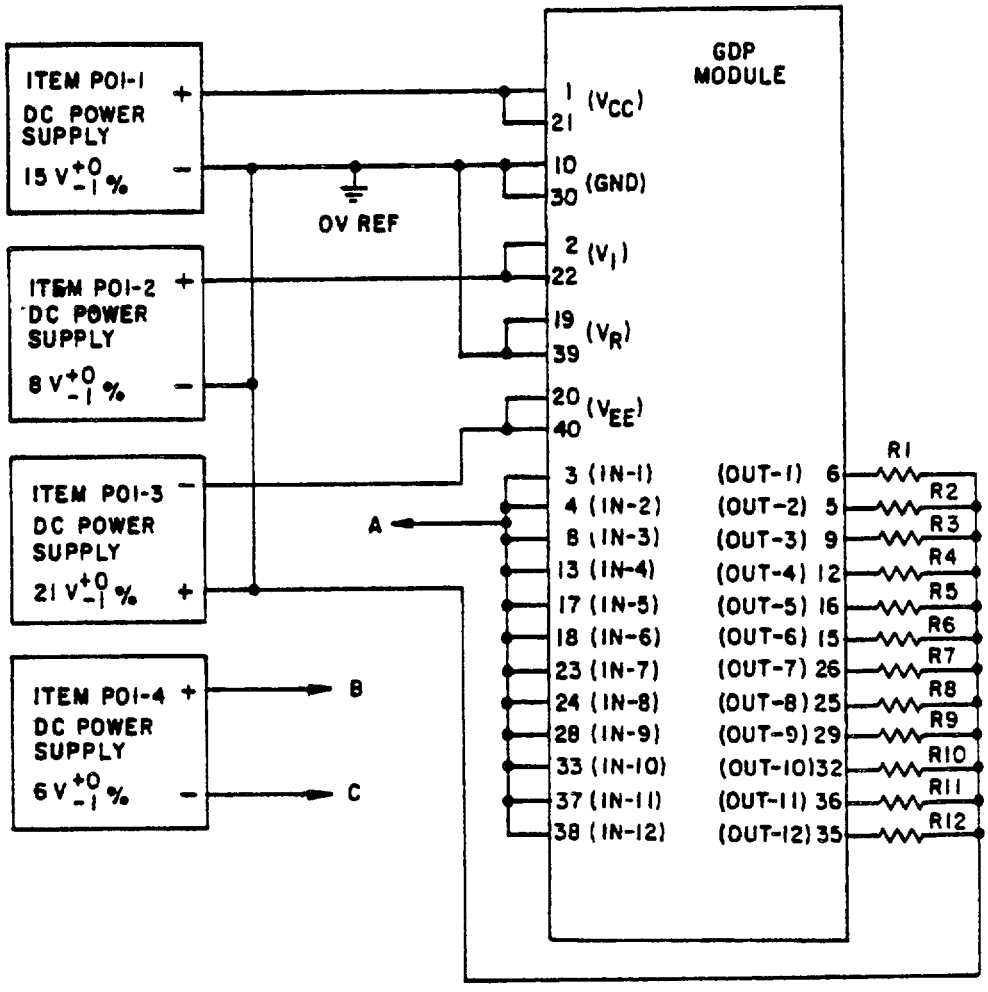
TABLE II. Stress (maximum voltage ratings) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
A	Stress (maximum voltage ratings)	B	A	10,30	--	--	--	4

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation	All pins	Frame	10	∞	MΩ	5
C2	"	All pins except 7,11, 14,27,31,34	7,11,14,27, 31,34	"	"	"	"
C3	"	1	21	"	"	"	"
C4	"	2	22	"	"	"	"
C5	"	19	39	"	"	"	"
C6	"	20	40	"	"	"	"
C7	"	10	30	"	"	"	"

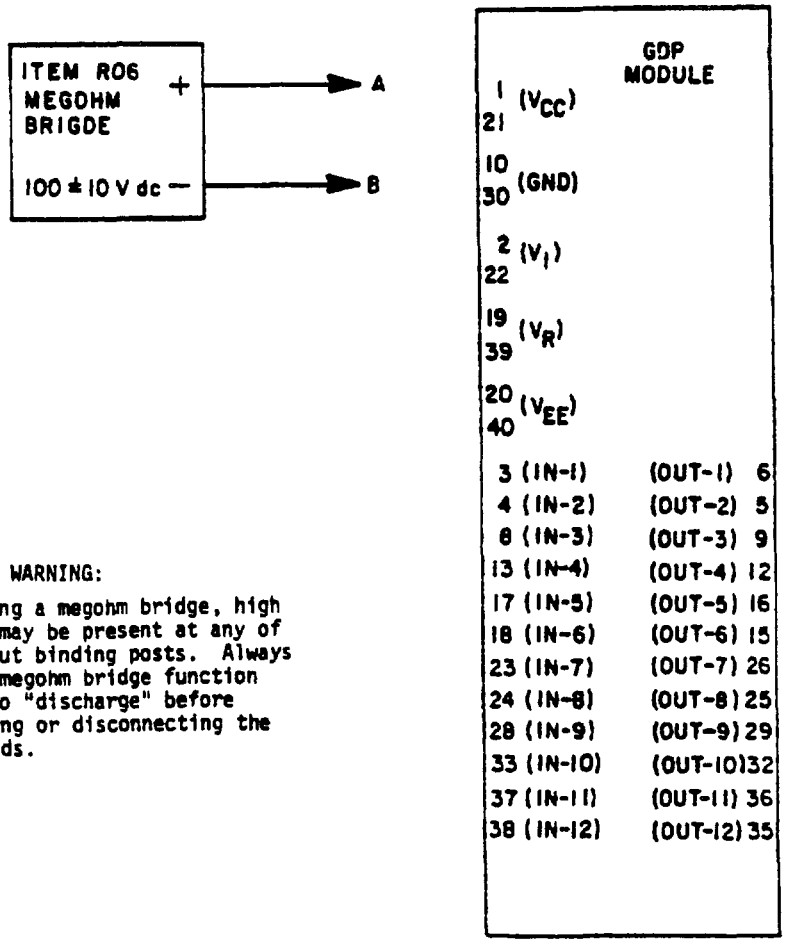
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NOTES:
1. R1-R12 = 3.57 kΩ ±1%, 1/8 W.
2. Make all circuit connections and apply voltage for a minimum of one minute. No electrical measurements shall be taken.

FIGURE 4. Stress (maximum voltage ratings) test circuit for table II.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition C1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input current (high-level and low-level) test.

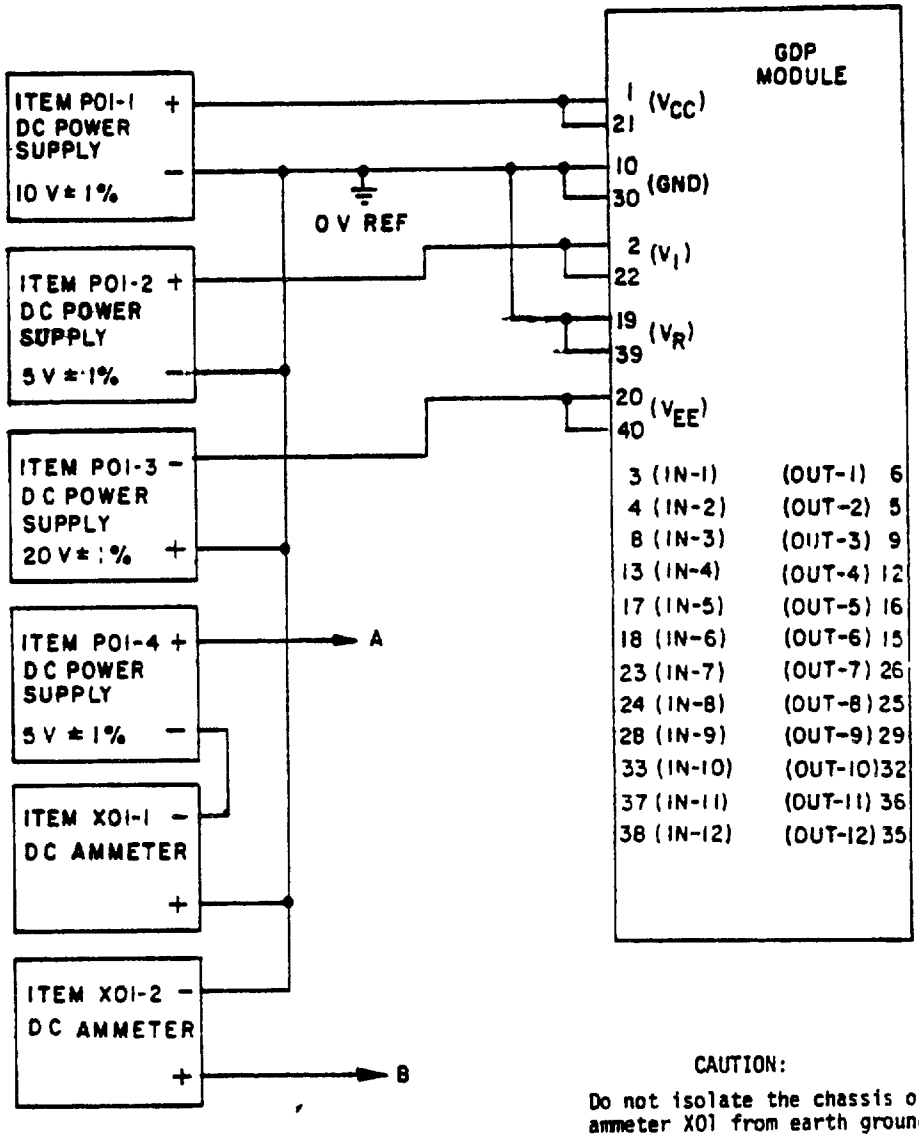
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	IN-1 Input current, high level	3	Open	-1	10	µA	6
D2	IN-2 "	4	"	"	"	"	"
D3	IN-3 "	8	"	"	"	"	"
D4	IN-4 "	13	"	"	"	"	"
D5	IN-5 "	17	"	"	"	"	"
D6	IN-6 "	18	"	"	"	"	"
D7	IN-7 "	23	"	"	"	"	"
D8	IN-8 "	24	"	"	"	"	"
D9	IN-9 "	28	"	"	"	"	"
D10	IN-10 "	33	"	"	"	"	"
D11	IN-11 "	37	"	"	"	"	"
D12	IN-12 "	38	"	"	"	"	"
E1	IN-1 Input current, low level	Open	3	0	1	mA	"
E2	IN-2 "	"	4	"	"	"	"
E3	IN-3 "	"	8	"	"	"	"
E4	IN-4 "	"	13	"	"	"	"
E5	IN-5 "	"	17	"	"	"	"
E6	IN-6 "	"	18	"	"	"	"
E7	IN-7 "	"	23	"	"	"	"
E8	IN-8 "	"	24	"	"	"	"
E9	IN-9 "	"	28	"	"	"	"

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TABLE IV. Input current (high-level and low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E10	IN-10 Input current, low level	Open	33	0	1	mA	6
E11	IN-11 Input current, low level	Open	37	0	1	mA	6
E12	IN-12 Input current, low level	Open	38	0	1	mA	6

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NOTE: Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 6. Input current (high-level and low-level) test circuit for table IV.

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TABLE V. Output voltage (high-level and low-level) test.

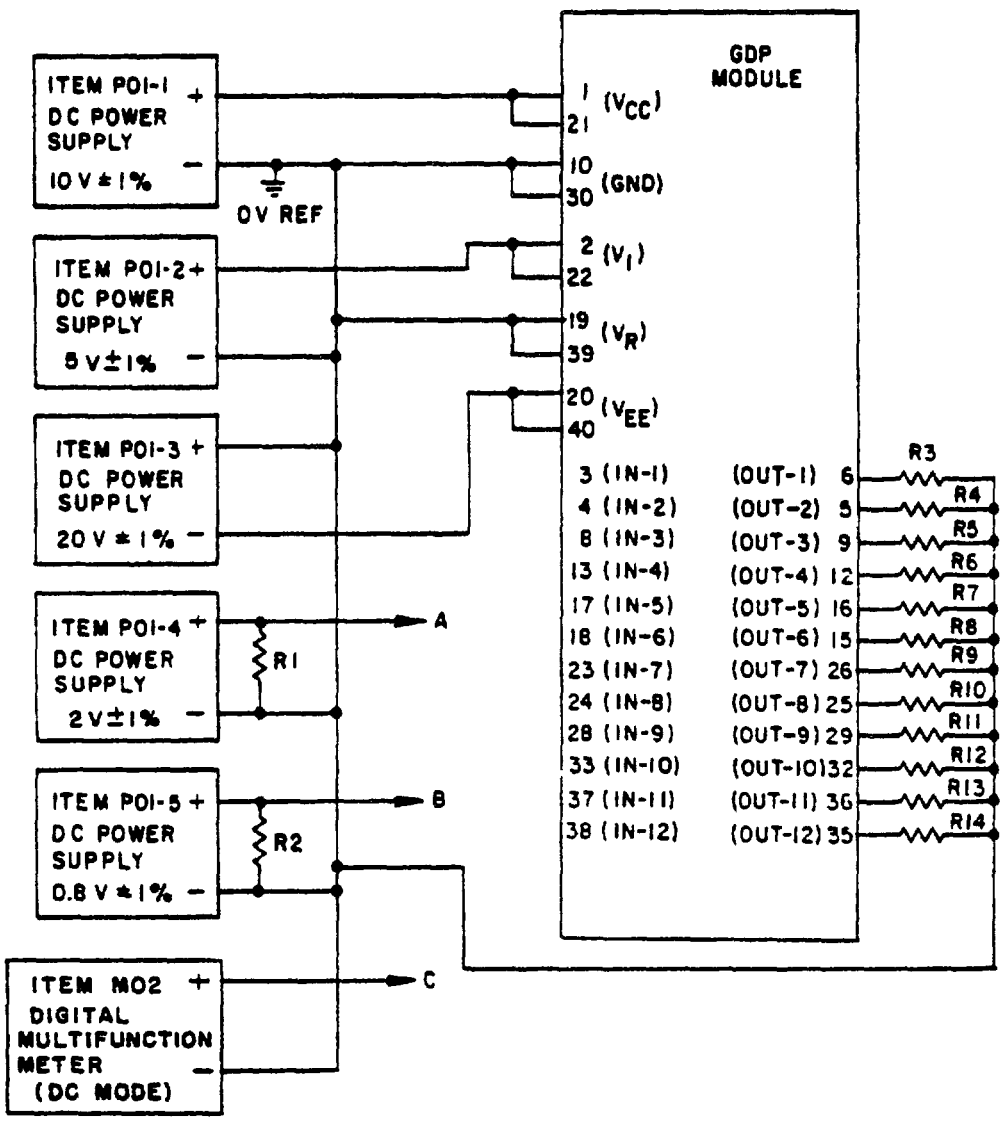
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	OUT-1 Output voltage, high level	3	Open	6	7.5	10.0	V	7
F2	OUT-2 "	4	"	5	"	"	"	"
F3	OUT-3 "	8	"	9	"	"	"	"
F4	OUT-4 "	13	"	12	"	"	"	"
F5	OUT-5 "	17	"	16	"	"	"	"
F6	OUT-6 "	18	"	15	"	"	"	"
F7	OUT-7 "	23	"	26	"	"	"	"
F8	OUT-8 "	24	"	25	"	"	"	"
F9	OUT-9 "	28	"	29	"	"	"	"
F10	OUT-10 "	33	"	32	"	"	"	"
F11	OUT-11 "	37	"	36	"	"	"	"
F12	OUT-12 "	38	"	35	"	"	"	"
G1	OUT-1 Output voltage, low level	Open	3	6	-17.5	-20.0	"	"
G2	OUT-2 "	"	4	5	"	"	"	"
G3	OUT-3 "	"	8	9	"	"	"	"
G4	OUT-4 "	"	13	12	"	"	"	"
G5	OUT-5 "	"	17	16	"	"	"	"
G6	OUT-6 "	"	18	15	"	"	"	"
G7	OUT-7 "	"	23	26	"	"	"	"
G8	OUT-8 "	"	24	25	"	"	"	"
G9	OUT-9 "	"	28	29	"	"	"	"

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TABLE V. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G10	OUT-10 Output voltage, low level	Open	33	32	-17.5	-20.0	V	7
G11	OUT-11 Output voltage, low level	Open	37	36	-17.5	-20.0	V	7
G12	OUT-12 Output voltage, low level	Open	38	35	-17.5	-20.0	V	7

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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. R3-R14 = 3.57 kΩ ±1%, 1/8 W.

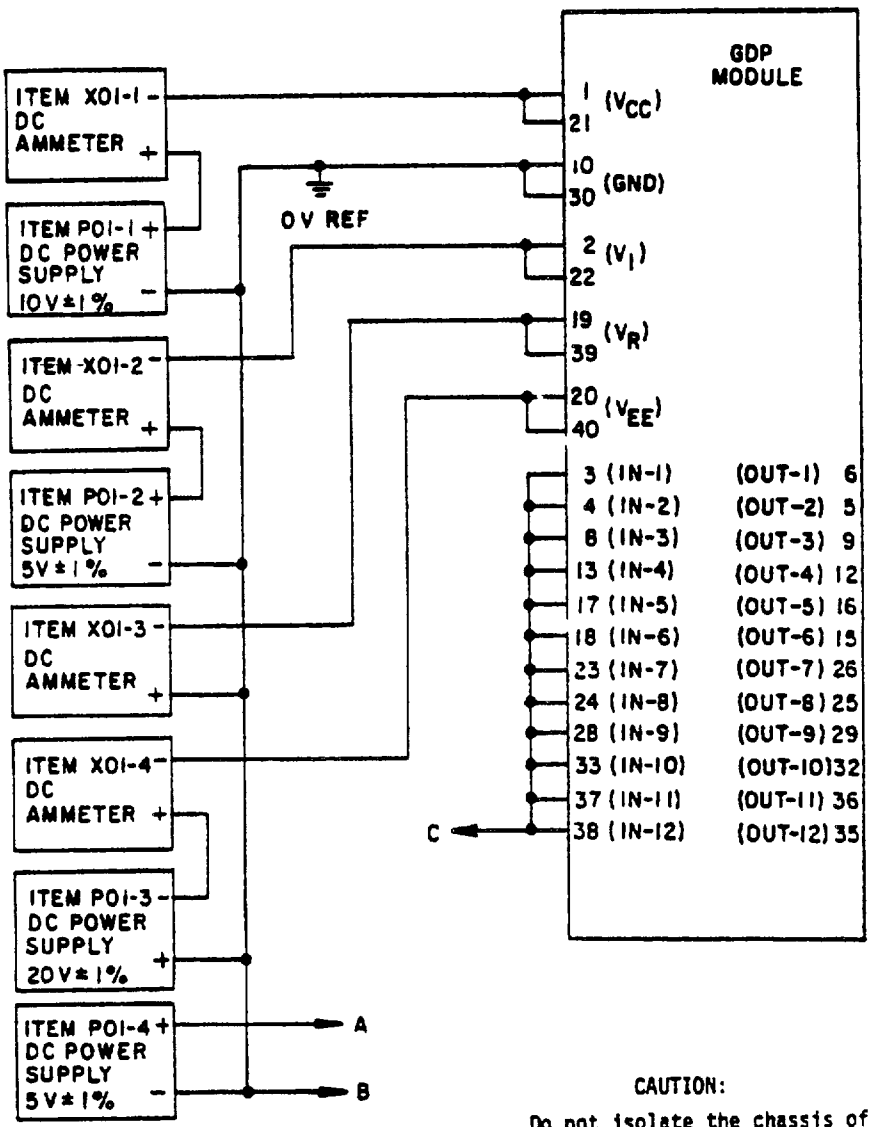
* FIGURE 7. Output voltage (high-level and low-level) test circuit for table V

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*TABLE VI. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Power supply current, high level I_{CC} (V_{CC} = 10 V)	C	Open	0	45	mA	8
H2	Power supply current, high level I_{EE} (V_{EE} = -20 V)	"	"	0	-24	"	"
H3	Power supply current, high level I_I (V_I = 5 V)	"	"	4	24	"	"
H4	Power supply current, high level I_R (V_R = 0 V)	"	"	-1	-54	"	"
J1	Power supply current, low level I_{CC} (V_{CC} = 10 V)	Open	C	0	3	"	"
J2	Power supply current, low level I_{EE} (V_{EE} = -20 V)	"	"	-3	-30	"	"
J3	Power supply current, low level I_I (V_I = 5 V)	"	"	3	36	"	"
J4	Power supply current, low level I_R (V_R = 0 V)	"	"	0.0	-9.6	"	"

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NOTE: Since items XO1-1 through XO1-4 are not used simultaneously, only one item XO1 is required.

FIGURE 8. Power supply current (high-level and low-level) test circuit for table VI.

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TABLE VII. Propagation delay time (t_{PLH} and t_{PHL}) test.

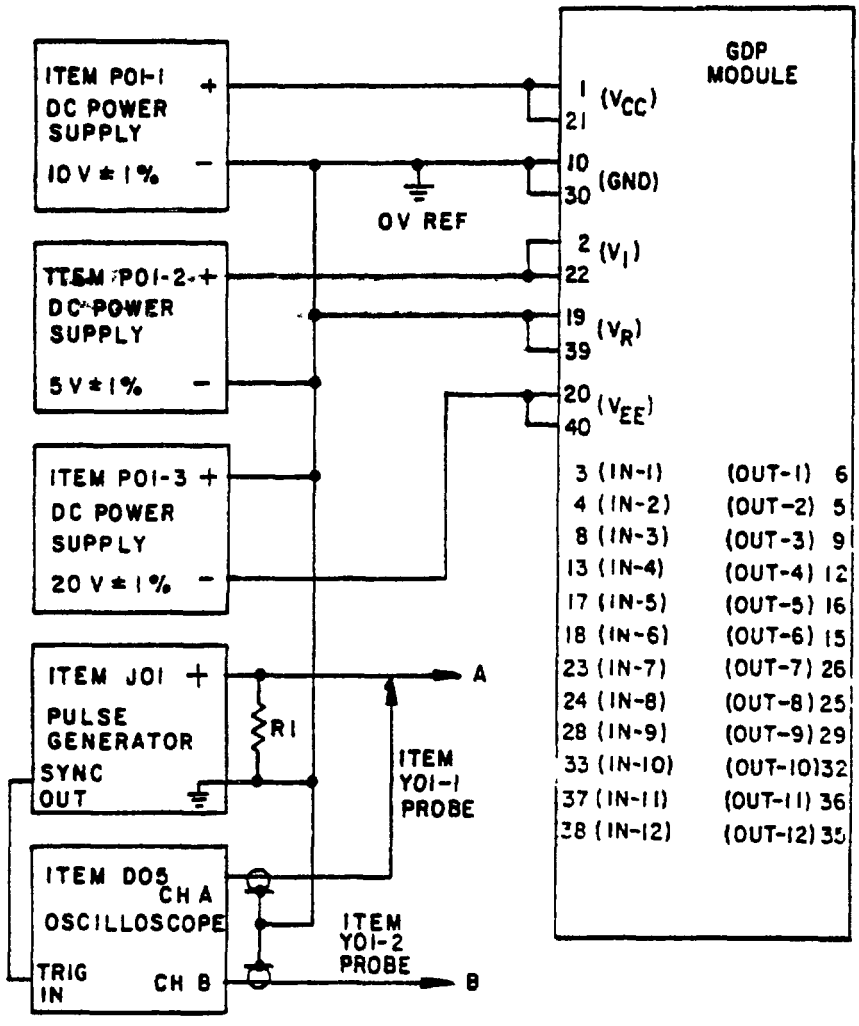
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	OUT-1 Propagation delay time (t_{PLH})	3	6	50	450	ns	9,10
K2	OUT-2 "	4	5	"	"	"	"
K3	OUT-3 "	8	9	"	"	"	"
K4	OUT-4 "	13	12	"	"	"	"
K5	OUT-5 "	17	16	"	"	"	"
K6	OUT-6 "	18	15	"	"	"	"
K7	OUT-7 "	23	26	"	"	"	"
K8	OUT-8 "	24	25	"	"	"	"
K9	OUT-9 "	28	29	"	"	"	"
K10	OUT-10 "	33	32	"	"	"	"
K11	OUT-11 "	37	36	"	"	"	"
K12	OUT-12 "	38	35	"	"	"	"
L1	OUT-1 Propagation delay time (t_{PHL})	3	6	"	"	"	"
L2	OUT-2 "	4	5	"	"	"	"
L3	OUT-3 "	8	9	"	"	"	"
L4	OUT-4 "	13	12	"	"	"	"
L5	OUT-5 "	17	16	"	"	"	"
L6	OUT-6 "	18	15	"	"	"	"
L7	OUT-7 "	23	26	"	"	"	"
L8	OUT-8 "	24	25	"	"	"	"
L9	OUT-9 "	28	29	"	"	"	"

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TABLE VII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L10	OUT-10 Propagation delay time (t_{PHL})	33	32	50	450	ns	9,10
L11	OUT-11 Propagation delay time (t_{PHL})	37	36	50	450	ns	9,10
L12	OUT-12 Propagation delay time (t_{PHL})	38	35	50	450	ns	9,10

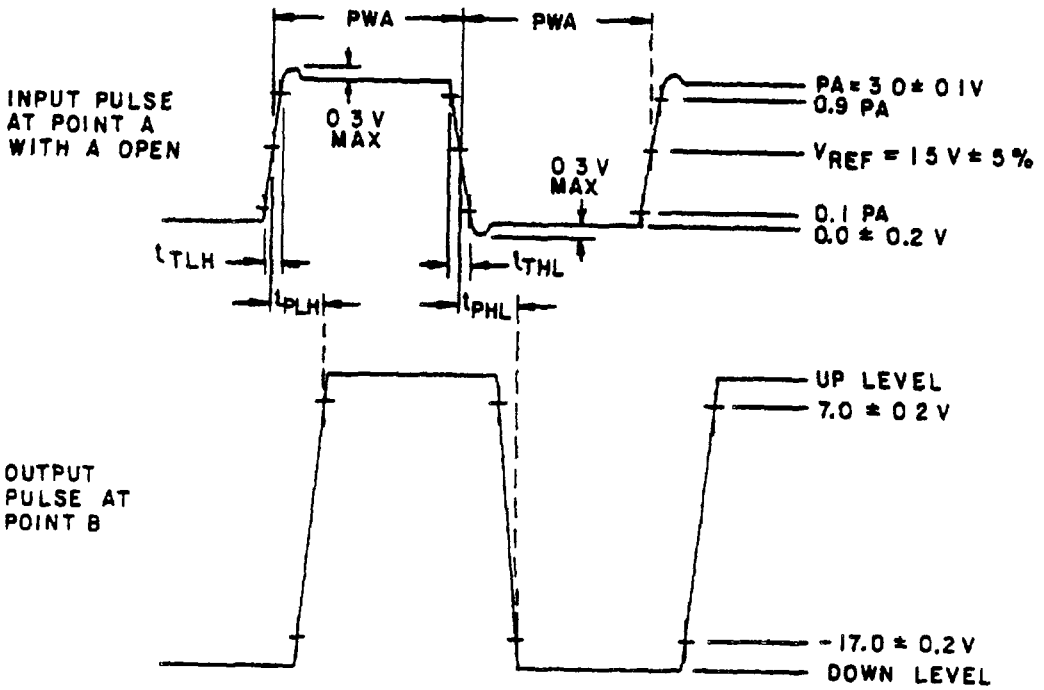
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. All outputs shall be loaded with R = 3.57 kΩ ±1%, 1/8 W, and C = 50 ±5 pF in parallel from the output to ground (capacitance includes probe and parasitic capacitance).

FIGURE 9. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table VII.

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NOTES.
1 Input pulse parameters
PWA = 1 0 ±0.1 us.
$t_{TLH} = t_{THL} = 25 \pm 5$ ns.
2 Before performing test, verify that the UP level is 7.5 V minimum and the DOWN level is -17.5 V minimum.

FIGURE 10. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table VII.

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*TABLE VIII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	OUT-1	Output crosstalk, low level	4,8,13,17, 18,23,24, 28,33,37, 38	6	Open	3	0	1	V	11,12
M2	OUT-2	"	3,8,13,17, 18,23,24, 28,33,37, 38	5	"	4	"	"	"	"
M3	OUT-3	"	3,4,13,17, 18,23,24, 28,33,37, 38	9	"	8	"	"	"	"
M4	OUT-4	"	3,4,8,17, 18,23,24, 28,33,37, 38	12	"	13	"	"	"	"
M5	OUT-5	"	3,4,8,13, 18,23,24, 28,33,37, 38	16	"	17	"	"	"	"
M6	OUT-6	"	3,4,8,13, 17,23,24, 28,33,37, 38	15	"	18	"	"	"	"
M7	OUT-7	"	3,4,8,13, 17,18,24, 28,33,37, 38	26	"	23	"	"	"	"
M8	OUT-8	"	3,4,8,13, 17,18,23, 28,33,37, 38	25	"	24	"	"	"	"

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*TABLE VIII. Output crosstalk (low-level and high-level) test - Continued.

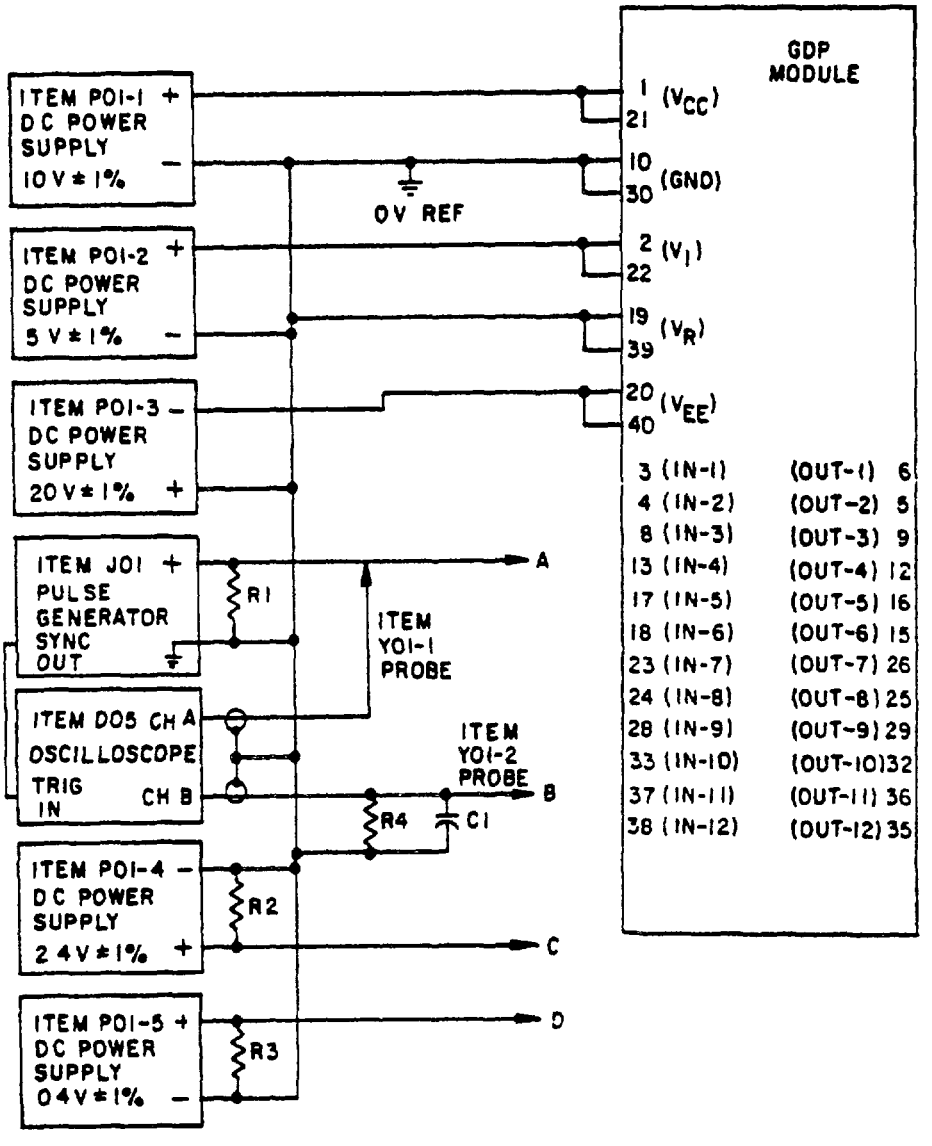
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M9	OUT-9	Output crosstalk, low level	3,4,8,13, 17,18,23, 24,33,37, 38	29	Open	28	0	1	V	11,12
M10	OUT-10	"	3,4,8,13, 17,18,23, 24,28,37, 38	32	"	33	"	"	"	"
M11	OUT-11	"	3,4,8,13, 17,18,23, 24,28,33, 38	36	"	37	"	"	"	"
M12	OUT-12	"	3,4,8,13, 17,18,23, 24,28,33, 37	12	13	Open	"	"	"	"
N1	OUT-1	Output crosstalk, high level	4,8,13,17, 18,23,24, 28,33,37, 38	6	3	"	"	800	mV	"
N2	OUT-2	"	3,8,13,17, 18,23,24, 28,33,37, 38	5	4	"	"	"	"	"
N3	OUT-3	"	3,4,13,17, 18,23,24, 28,33,37, 38	9	8	"	"	"	"	"
N4	OUT-4	"	3,4,8,17, 18,23,24, 28,33,37, 38	12	13	"	"	"	"	"

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*TABLE VIII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N5	OUT-5 Output crosstalk, high level	3,4,8,13, 23,24,28, 33,37,38	16	17	Open	0	800	mV	11,12
N6	OUT-6 "	3,4,8,13, 17,23,24, 28,33,37, 38	15	18	"	"	"	"	"
N7	OUT-7 "	3,4,8,13, 17,18,24, 28,33,37, 38	26	23	"	"	"	"	"
N8	OUT-8 "	3,4,8,13, 17,18,23, 28,33,37, 38	25	24	"	"	"	"	"
N9	OUT-9 "	3,4,8,13, 17,18,23, 24,33,37, 38	29	28	"	"	"	"	"
N10	OUT-10 "	3,4,8,13, 17,18,23, 24,28,37, 38	32	33	"	"	"	"	"
N11	OUT-11 "	3,4,8,13, 17,18,23, 24,28,33, 38	36	37	"	"	"	"	"
N12	OUT-12 "	3,4,8,13, 17,18,23, 24,28,33, 37	35	38	"	"	"	"	"

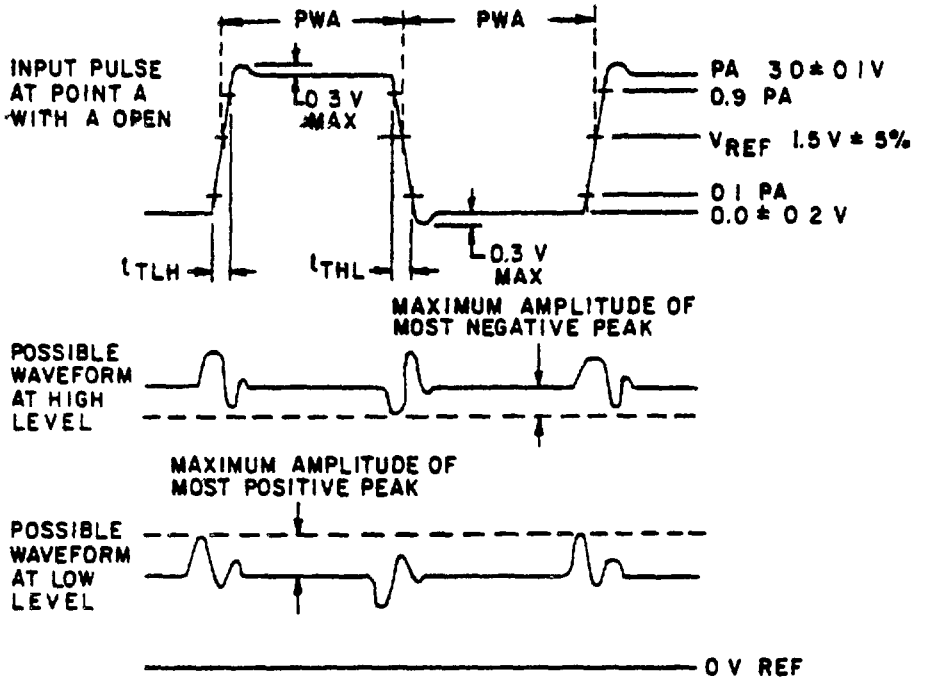
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NOTES:
1. R1-R3 = 50Ω ±5%, 1 W.
2. R4 = 3.57 kΩ ±1%, 1/8 W.
3. C1 = 30 ±5 pF.

FIGURE 11. Output crosstalk (low-level and high-level) test circuit for table VIII.

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NOTES:
1. Input pulse parameters:
   PWA = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL} = 15 \pm 2$ ns.
2. Capacitive filtering may be required on the V_{CC} and V_{EE} power applies. Each supply shall have a maximum of 400 mV ripple.

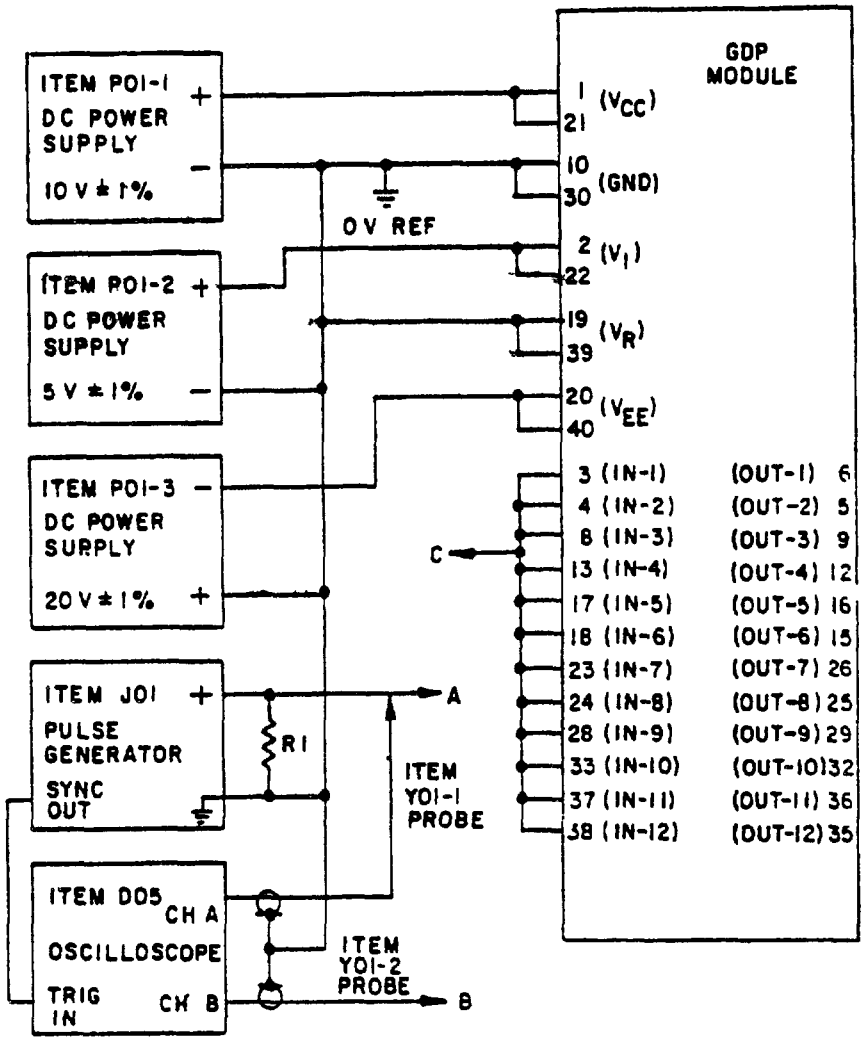
FIGURE 12. Output crosstalk (low-level and high-level) test circuit waveforms for table VIII.

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TABLE IX. Power supply transient amplitude test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Power supply transient amplitude (V_1)	C	2,22	0	400	mV	13,14
P2	Power supply transient amplitude (V_{CC})	C	1,21	0	2	V	13,14
P3	Power supply transient amplitude (V_{EE})	C	20,40	0	2	V	13,14

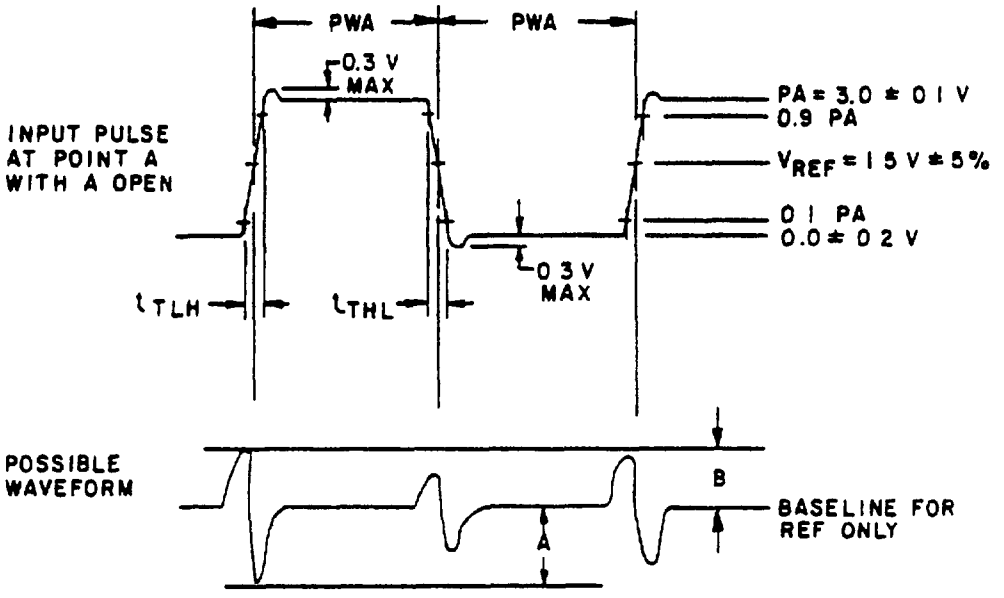
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NOTE: R1 = 50Ω ±5%, 1 W.

FIGURE 13. Power supply transient amplitude test circuit for table IX

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NOTES
1. Input pulse parameters.
   PWA = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
2. For test conditions P1 and P2, the value is measured from baseline to the most negative excursion of the waveform.
3. For test condition P3, the value is measured from baseline to the most positive excursion of the waveform.

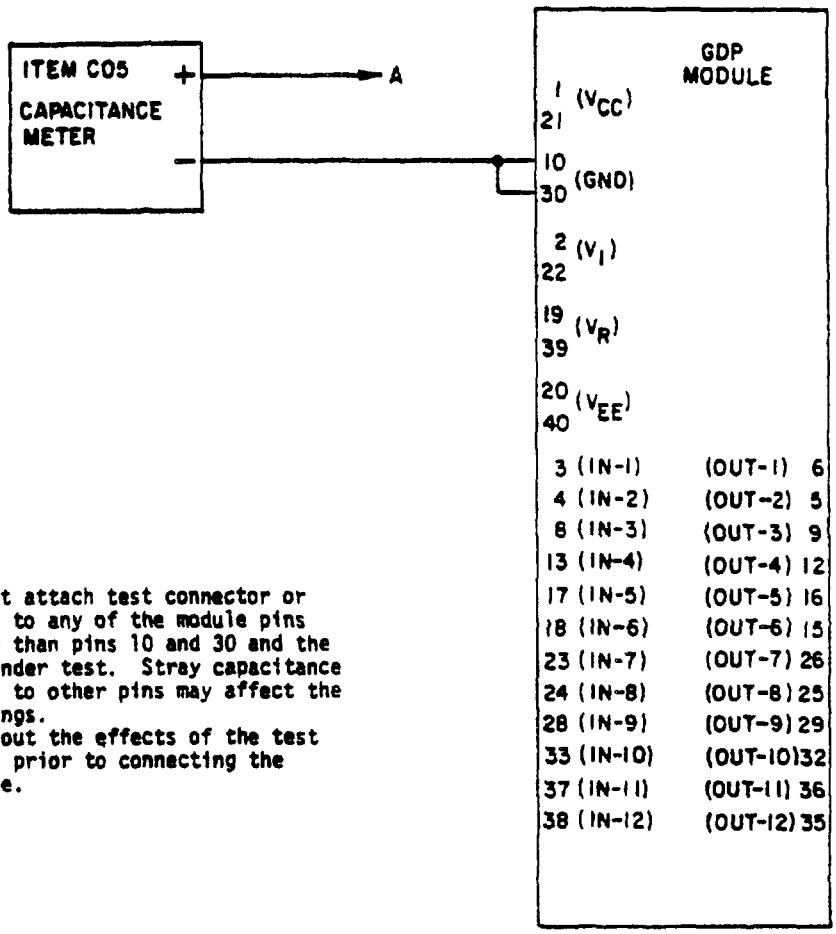
FIGURE 14. Power supply transient amplitude test circuit waveforms for table IX.

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TABLE X. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
Q1	IN-1 Input capacitance	3	2	15	pF	15
Q2	IN-2 "	4	"	"	"	"
Q3	IN-3 "	8	"	"	"	"
Q4	IN-4 "	13	"	"	"	"
Q5	IN-5 "	17	"	"	"	"
Q6	IN-6 "	18	"	"	"	"
Q7	IN-7 "	23	"	"	"	"
Q8	IN-8 "	24	"	"	"	"
Q9	IN-9 "	28	"	"	"	"
Q10	IN-10 "	33	"	"	"	"
Q11	IN-11 "	37	"	"	"	"
Q12	IN-12 "	38	"	"	"	"

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pins 10 and 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 15. Input capacitance test circuit for table X.

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